Operating Systems

Week 1 – Introduction

Korea University Sejong Campus, Department of Computer Science & Software

Part 1. OT

Course Orientation

Instructor

Unggi Lee, codingchild@korea.ac.kr

- Assistant Professor, Dept. of Computer Science & Software, Korea University Sejong Campus
- Previously:
  - Assistant Professor, Dept. of Computer Engineering, Chosun University
  - AI/NLP Engineer in Global EdTech (Enuma, I-Scream Edu)
  - Elementary School Teacher
- Research (Google Scholar):
  - AIED: Generative AI in Education, Pedagogical Alignment, Knowledge Tracing
  - NLP & Robotics: Large Language Models (LLMs), Vision-Language-Action (VLA)
- Lab Activities (Lab):
  - Published a VLA preprint with undergraduates at Chosun University
  - Industry–academia partnerships with Upstage, Neudive, and others
  - Collaborations with researchers in the US, Singapore, and the UK

Syllabus

- You can find the syllabus in **LMS**
- Total **15 weeks**
  - 8th week – **Midterm Exam**
  - 15th week – **Final Exam**

Grading

Component	Weight
Assignment	**10%**
Midterm Exam (written)	**30%**
Final Exam (written)	**30%**
Final Exam (project)	**30%**
Attendance	0%

However, absent for **one-third (1/3)** or more of the total class hours → no grade will be awarded.

Assignment (10%)

In-class Quiz: 5%

- Quizzes in **10 classes** → each **0.5%**
- Week 3, 4, 5, 6, 7, 9, 10, 11, 12, 13

Take-home Assignment: 5%

- Assignments in **5 classes** → each **1%**
- Week 2, 3, 4, 5, 6

Midterm & Final Exam (Written)

- **Handwritten** (digital devices not allowed)
- **1 hour** each

Final Exam – Project (30%)

- Starting from **Week 9**
- **3–4 members** per team
- Unlimited use of **coding agents**
  - Claude Code, Codex, Gemini CLI, OpenCode, etc.
- You must:
  - **Design and develop** an OS prototype
    - e.g., Upgrade the existing xv6 with new features
    - e.g., Design an LLM-powered OS concept
  - Write **OS spec documentation** and a **project report**
  - Give a **presentation** (in-class, Week 14)
- Evaluation:
  - Professor: **15%**
  - Students (peer review): **15%**

Class Format

Hour	Content
1st	Theory Lecture (Part 1) + Quiz
2nd	Theory Lecture (Part 2)
3rd	Hands-on Lab

- Textbook: Silberschatz, **Operating System Concepts** 10th Edition
- Lab reference: **xv6** (RISC-V), MIT 6.1810

Part 2. What is an Operating System?

A Big-Picture Overview

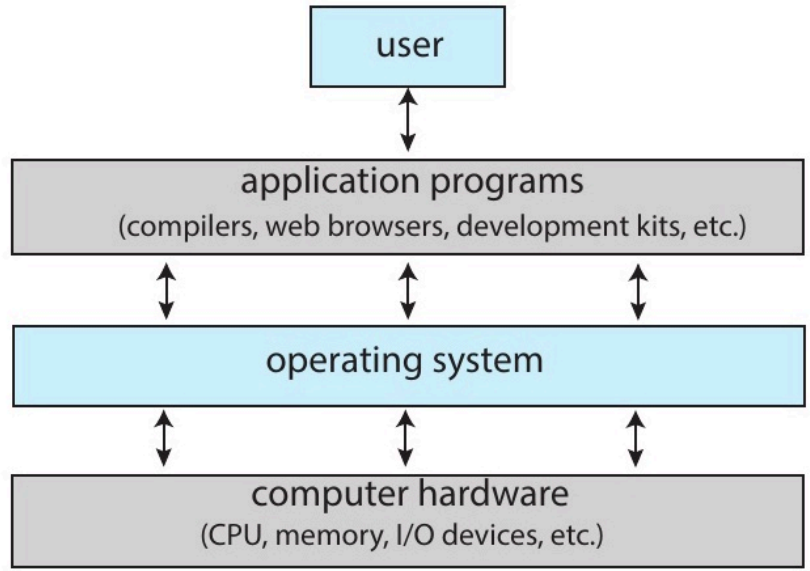
What is an Operating System?

- The **one program running at all times** on the computer = **Kernel**
- Everything else is either a **system program** or an **application program**

OS is like a government:

> It does not perform useful functions by itself, but provides an **environment** in which other programs can do useful work.

Where the OS Sits



Silberschatz, Figure 1.1 – Abstract view of the components of a computer system

- The OS sits **between** hardware and applications
- It directly controls hardware and provides a **clean interface** to programs

Two Roles of the OS

Resource Allocator

Manages and distributes **limited resources** efficiently and fairly:

- **CPU time** – which process runs when
- **Memory space** – allocation to each process
- **Storage & I/O** – disk access, device sharing

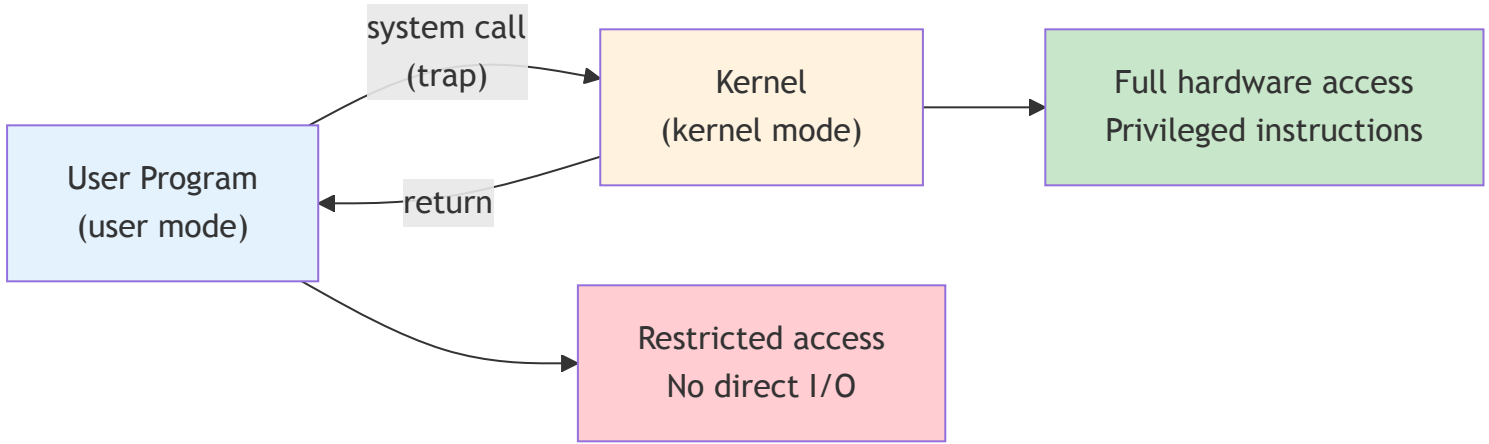
Control Program

Controls execution of user programs, prevents errors and misuse:

- **Manage execution** – start, stop, schedule programs
- **Prevent errors** – catch illegal operations
- **Enforce isolation** – protect processes from each other

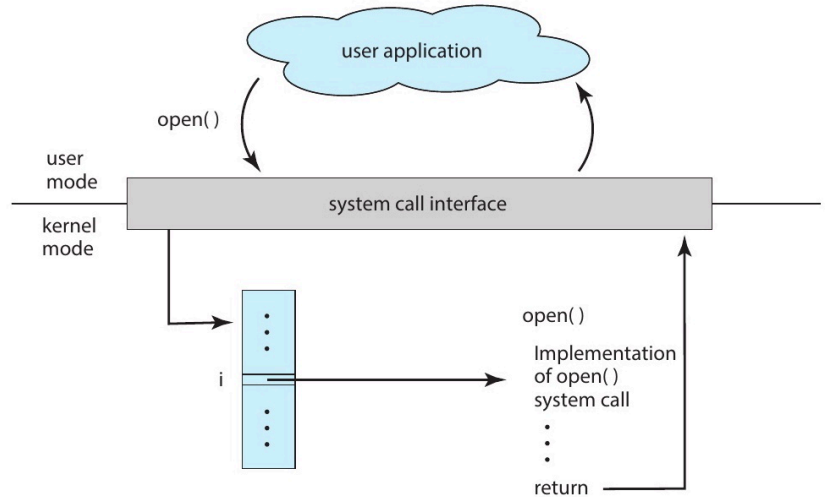
How Does the OS Protect Itself?

Dual-Mode Operation



Mode	Mode bit	Who runs here
Kernel mode	0	OS kernel – full hardware access
User mode	1	Applications – restricted access

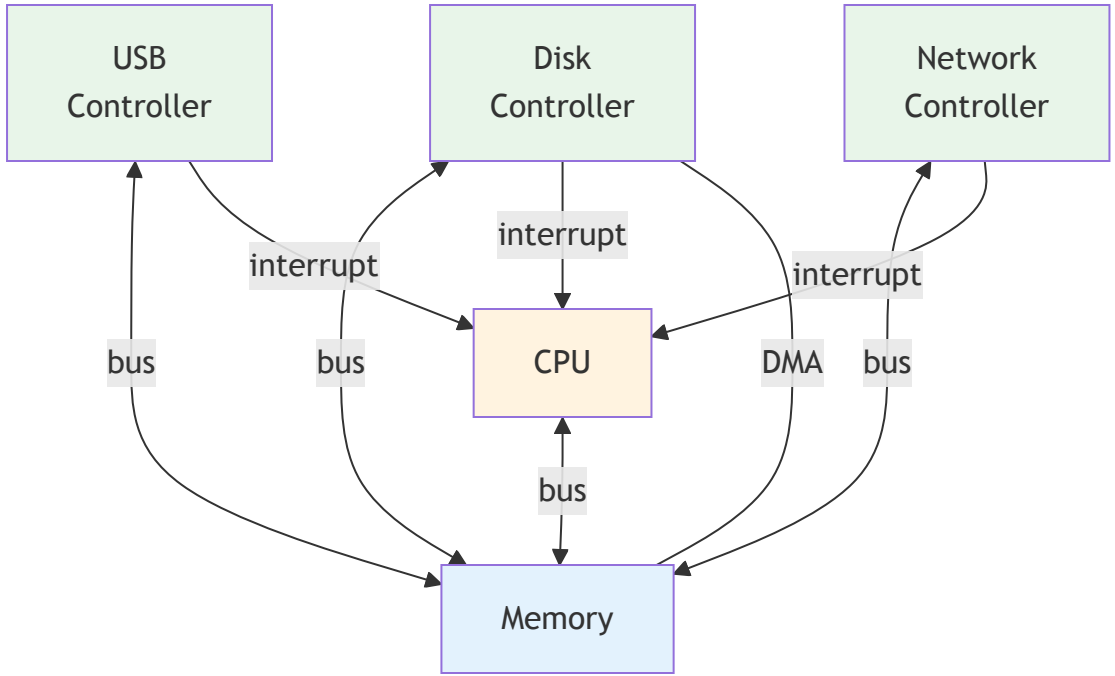
System Calls – The OS API



Silberschatz, Figure 2.6 – The handling of a user application invoking the open() system call

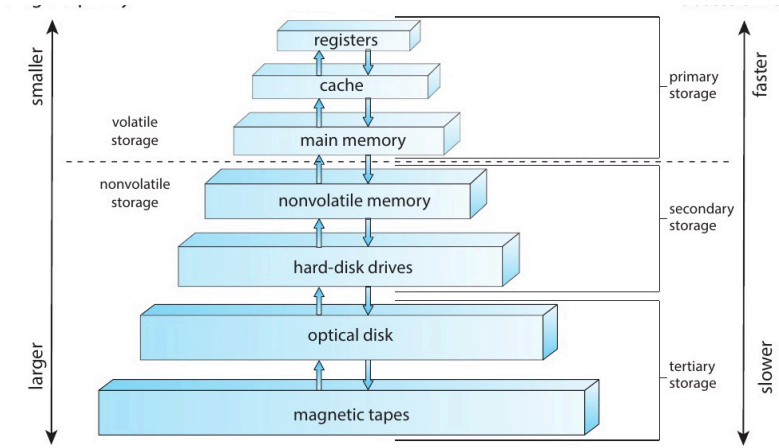
- **System call** = the only way for user programs to request OS services
- User program → C library → `syscall` instruction (trap) → kernel handles it → return
- Even a simple `cp in.txt out.txt` triggers **thousands** of system calls

How a Computer System Works



- Devices signal completion via **interrupts**
- **DMA** (Direct Memory Access): bulk data transfer without CPU intervention

Storage Hierarchy



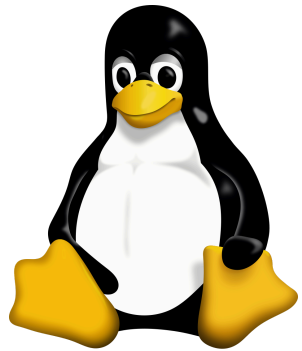
Silberschatz, Figure 1.6 – Storage-device hierarchy

Level	Size	Access Time	Managed by
Registers	< 1 KB	~0.3 ns	Hardware
Cache (L1/L2)	< 64 MB	~1–25 ns	Hardware

OS Structures

Structure	Key Idea	Example
Monolithic	Everything in one kernel binary	Linux, traditional UNIX
Microkernel	Minimal kernel + user-space services	Mach, QNX
Hybrid	Mix of both	macOS (Mach + BSD), Windows
Loadable Modules	Core kernel + dynamic modules	Linux (LKM)

Most modern OSes are **hybrid** – pragmatic, not pure.



What We'll Use: xv6

- **xv6**: A simple, Unix-like teaching OS by MIT
- Written in **C** for **RISC-V** architecture
- ~10,000 lines of code – small enough to read entirely
- Implements: processes, virtual memory, file system, shell
- We'll **read, modify, and extend** xv6 throughout the semester

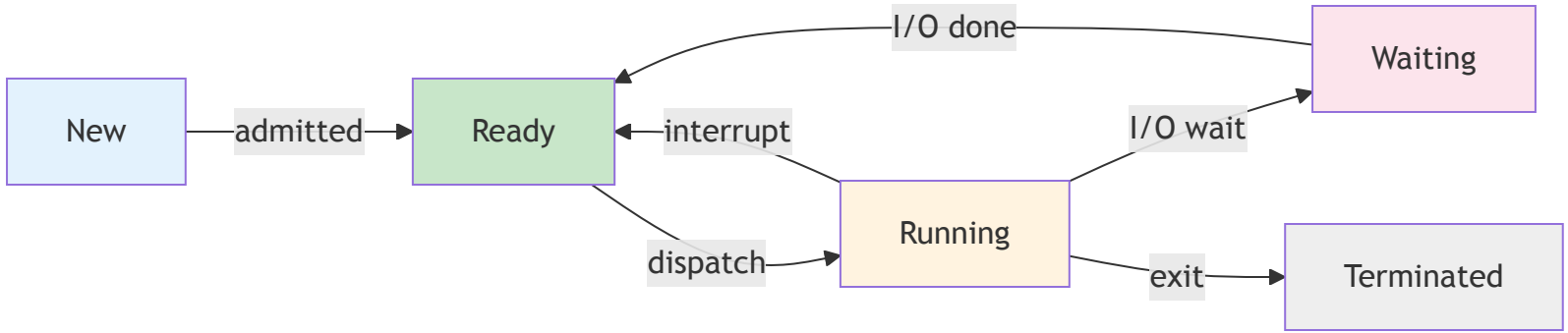
```
git clone https://github.com/mit-pdos/xv6-riscv
cd xv6-riscv
make qemu    # Boot xv6 in QEMU emulator
```

Part 3. Semester Preview

A taste of what's coming in each topic area

Preview: Process (Week 2–3)

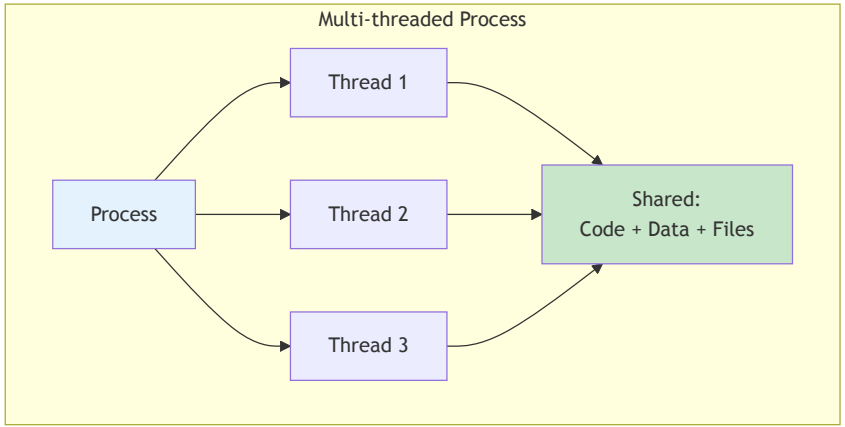
A **process** = a program in execution, with its own memory and state

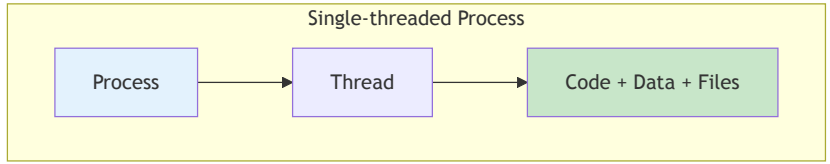


- Key system calls: `fork()`, `exec()`, `wait()`, `exit()`
- `fork()` creates a **copy** of the current process (parent → child)
- `exec()` **replaces** the process image with a new program
- In xv6: `kernel/proc.c` – the process table and state transitions

Preview: Threads & Concurrency (Week 4–5)

A **thread** = a lightweight execution unit that shares its process's address space

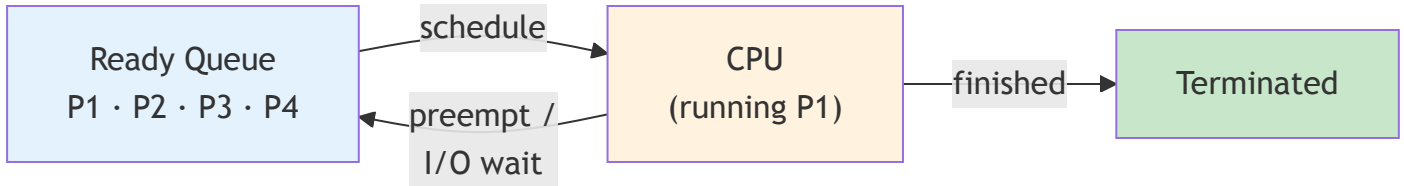




- Multiple threads in one process → **parallelism** on multi-core CPUs
- Challenge: **race conditions** when threads access shared data simultaneously

Preview: CPU Scheduling (Week 6–7)

The OS decides **which process runs next** on the CPU

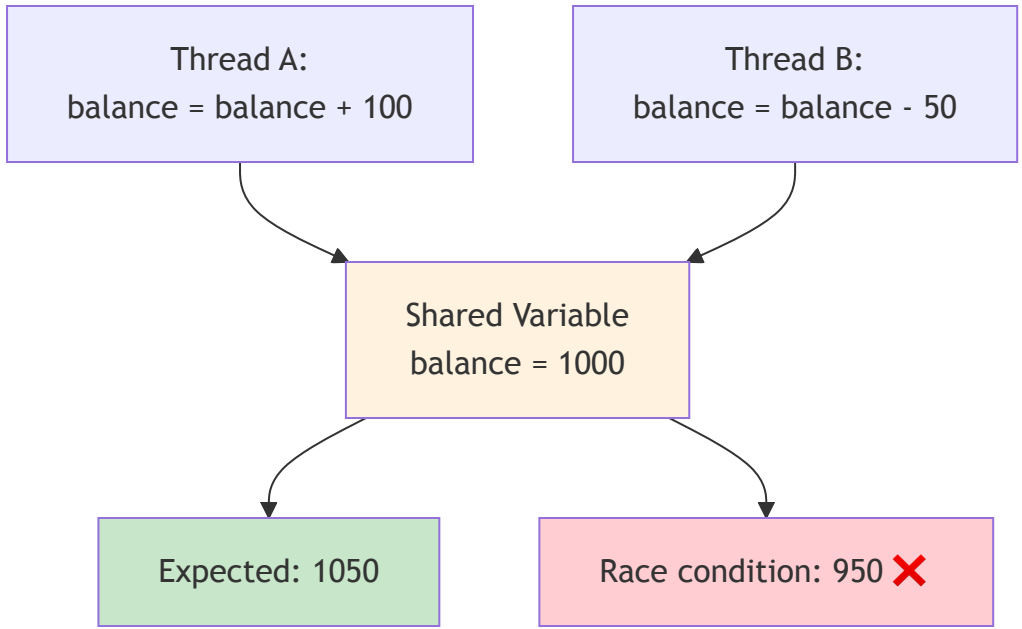


Scheduling algorithms balance competing goals:

Algorithm	Idea	Trade-off
FCFS	First come, first served	Simple but convoy effect
SJF	Shortest job first	Optimal avg wait, hard to predict
Round Robin	Fixed time slice, rotate	Fair, good response time

Preview: Synchronization (Week 9)

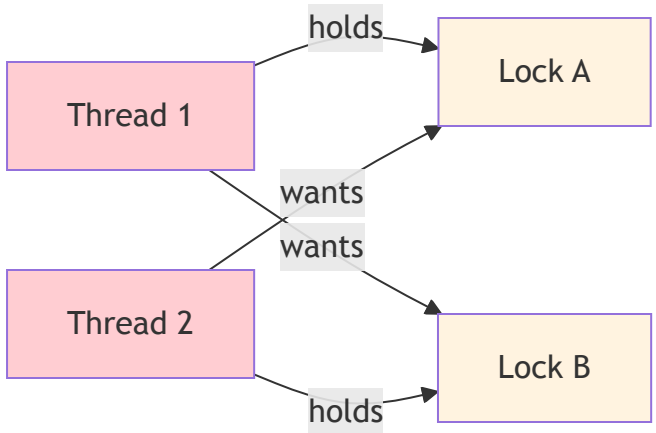
When multiple threads share data, we need **coordination**



- **Critical section**: code that accesses shared resources
- **Locks** (mutexes): ensure only one thread enters at a time

Preview: Deadlocks (Week 10)

Deadlock = two or more processes each waiting for the other to release a resource

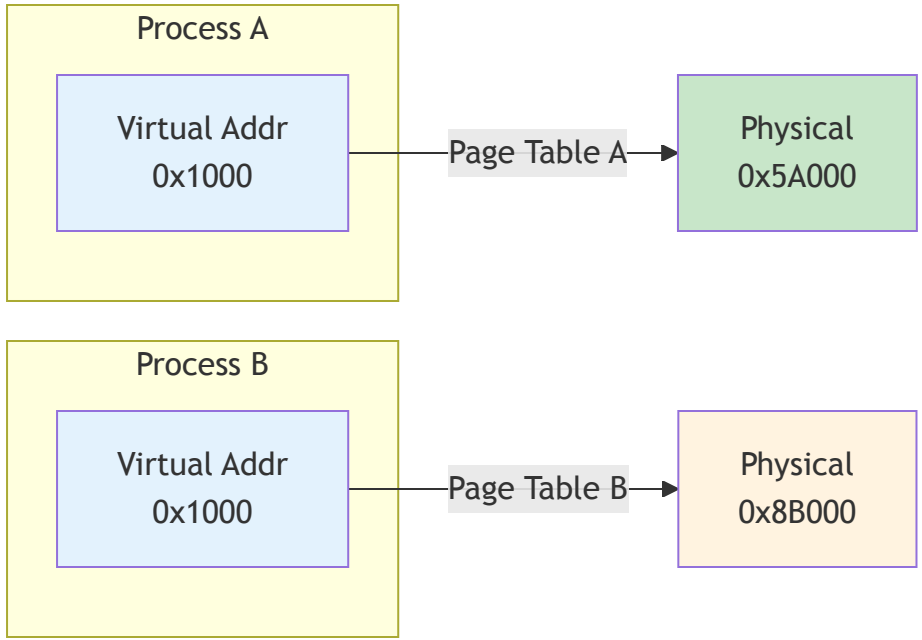


Four conditions (all must hold): Mutual exclusion · Hold and wait · No preemption · **Circular wait**

Solutions: Lock ordering · `trylock` + back-off · Deadlock detection & recovery

Preview: Memory Management (Week 11–12)

Virtual memory gives each process its own private address space



- Same virtual address → **different** physical locations (isolation!)
- **Page table**: per-process mapping from virtual pages to physical frames

Preview: File System & Security (Week 13–14)

File system organizes persistent data in layers

Layer	Component	Key Functions
6	**File Descriptors**	`open()`, `read()`, `write()`, `close()`
5	**Pathnames**	`namei()` – resolve `/path/to/file`
4	**Directories**	`dirlookup()`, `dirlink()`
3	**Inodes**	`ialloc()`, `readi()`, `writei()`
2	**Logging**	Crash safety via write-ahead log
1	**Buffer Cache**	`bread()`, `bwrite()` – cache disk blocks
0	**Disk**	Physical block device

Course Roadmap

Week	Topic	Week	Topic
1	Introduction + Coding Agents	**9**	Synchronization
2–3	Process	**10**	Deadlocks
4–5	Thread & Concurrency	**11–12**	Memory Management
6–7	CPU Scheduling	**13**	File System
8	*Midterm Exam*	**14**	Security + Project Presentation
		15	*Final Exam (Written)*

Summary

- **OS = Kernel**: The always-running program that manages hardware resources
- **Dual-mode** (user / kernel) protects the OS from user programs
- **System calls** are the interface between applications and the kernel
- Key topics: **Process, Thread, Scheduling, Sync, Memory, File System**
- We'll use **xv6** to see how a real OS works inside
- Next week: **Process** – fork, exec, wait, pipe

Q & A

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